

TE Curve Verification Pipeline Best Model Collage Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative TE Curve Verification Pipeline TE-curve predictions for the current best reference, RCIM Model-Bank Reproduction, Wave 1 directional, and Wave 1 global models. Each model is shown as one four-image collage so local oscillation tracking can be inspected directly.

Scope

- each collage contains four deterministic held-out test curves;
- forward models are shown on forward curves only;
- backward models are shown on backward curves only;
- global Wave 1 models are shown on two forward and two backward curves;
- Measured TE uses the same line width as predictions and a dark-gray color for balanced visual comparison.

Metrics Summary

Forward Accepted Non Windowed Reference Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
accepted_periodic_mlp_harmonic_fw	accepted_non_windowed_reference	Fw	0.001694	0.002008	3.439

Forward Accepted Time Windowed Incumbent Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
accepted_periodic_gru_sequence_fw	accepted_time_windowed_incumbent	Fw	0.001618	0.001931	3.278

Forward Frozen Analytical Anchor Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_stage15_pf_a_setpoint_quadratic_fw	frozen_analytical_anchor	Fw	0.001812	0.002126	3.749

Forward Wave52r Stage10 Non Temporal Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_stage10_r00	wave52r_stage10_non_temporal	Fw	0.001658	0.001986	3.422
wave52r_stage10_s01	wave52r_stage10_non_temporal	Fw	0.001658	0.001988	3.422

Forward Wave52r Stage12 Temporal Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_stage12_f01	wave52r_stage12_temporal	Fw	0.001444	0.001724	2.993
wave52r_stage12_s01	wave52r_stage12_temporal	Fw	0.001495	0.001768	3.041

Forward Wave52r Stage5 Non Temporal Models

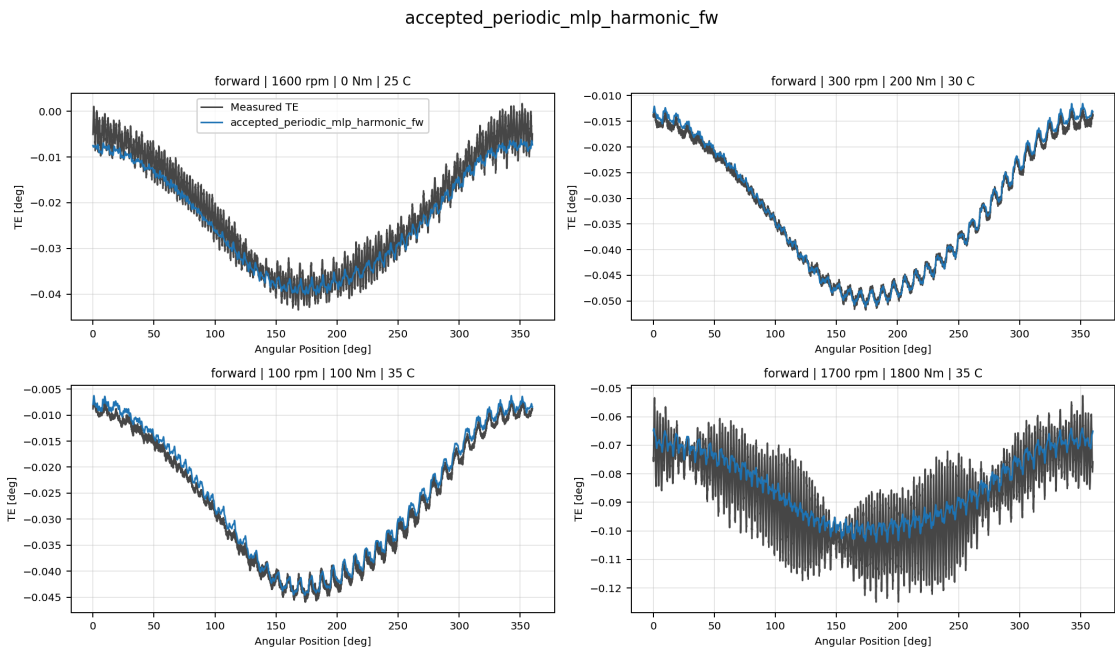
Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_stage5_h08_seed_314159	wave52r_stage5_non_temporal	Fw	0.001694	0.002005	3.483
wave52r_stage5_h04_seed_314159	wave52r_stage5_non_temporal	Fw	0.001728	0.002036	3.558

Forward Wave52r Stage9 Temporal Models

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52r_stage9_k01	wave52r_stage9_temporal	Fw	0.001374	0.001645	2.716

Collage Gallery - Forward Accepted Non Windowed Reference Models

accepted_periodic_mlp_harmonic_fw

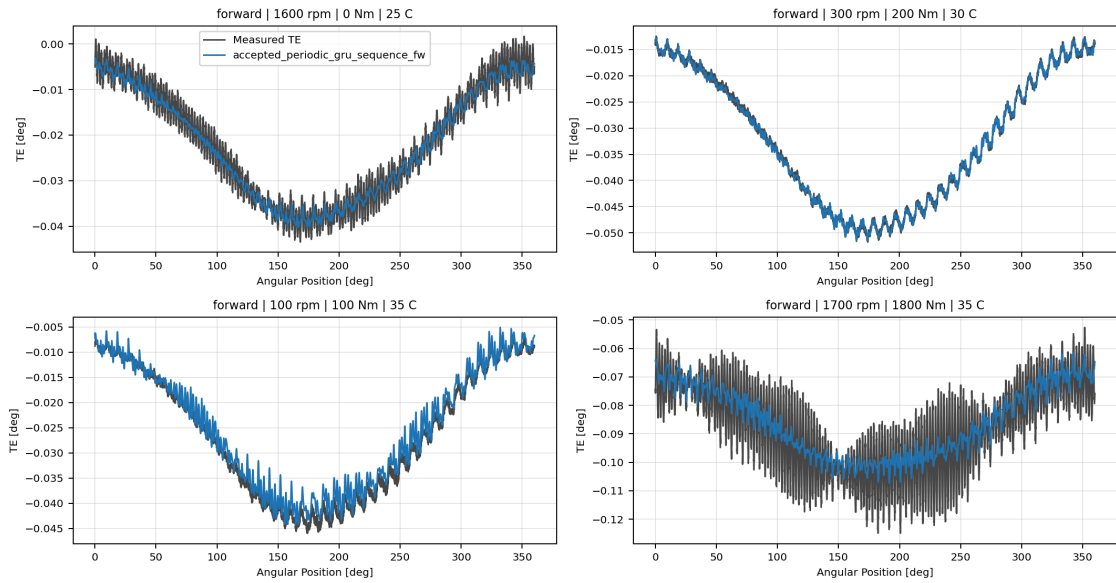


accepted_periodic_mlp_harmonic_fw TE Curve Verification Pipeline collage

Collage Gallery - Forward Accepted Time Windowed Incumbent Models

accepted_periodic_gru_sequence_fw

accepted_periodic_gru_sequence_fw

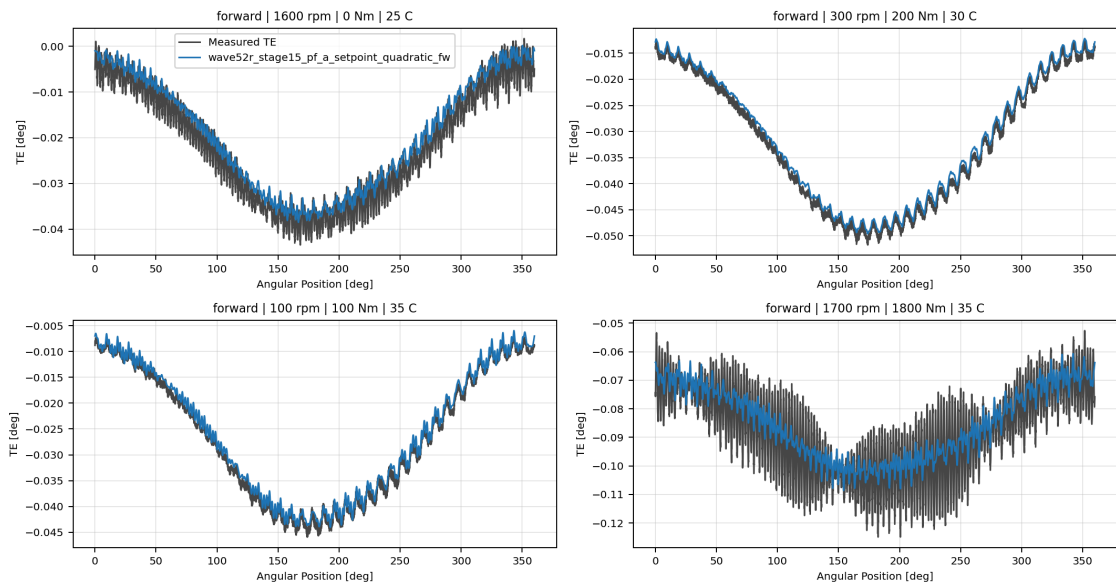


accepted_periodic_gru_sequence_fw TE Curve Verification Pipeline collage

Collage Gallery - Forward Frozen Analytical Anchor Models

wave52r_stage15_pf_a_setpoint_quadratic_fw

wave52r_stage15_pf_a_setpoint_quadratic_fw

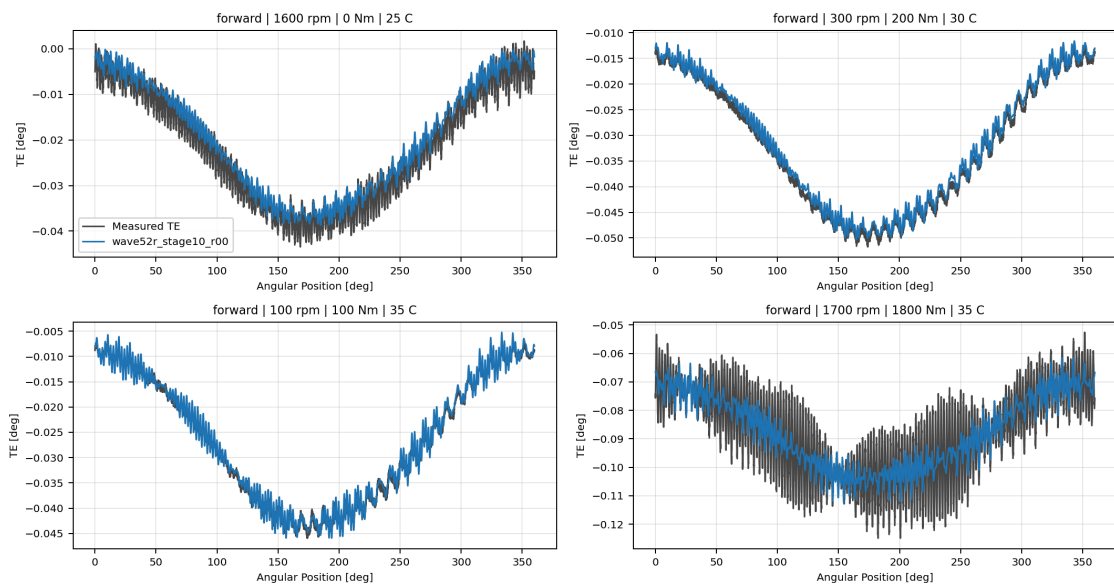


wave52r_stage15_pf_a_setpoint_quadratic_fw TE Curve Verification Pipeline collage

Collage Gallery - Forward Wave52r Stage10 Non Temporal Models

wave52r_stage10_r00

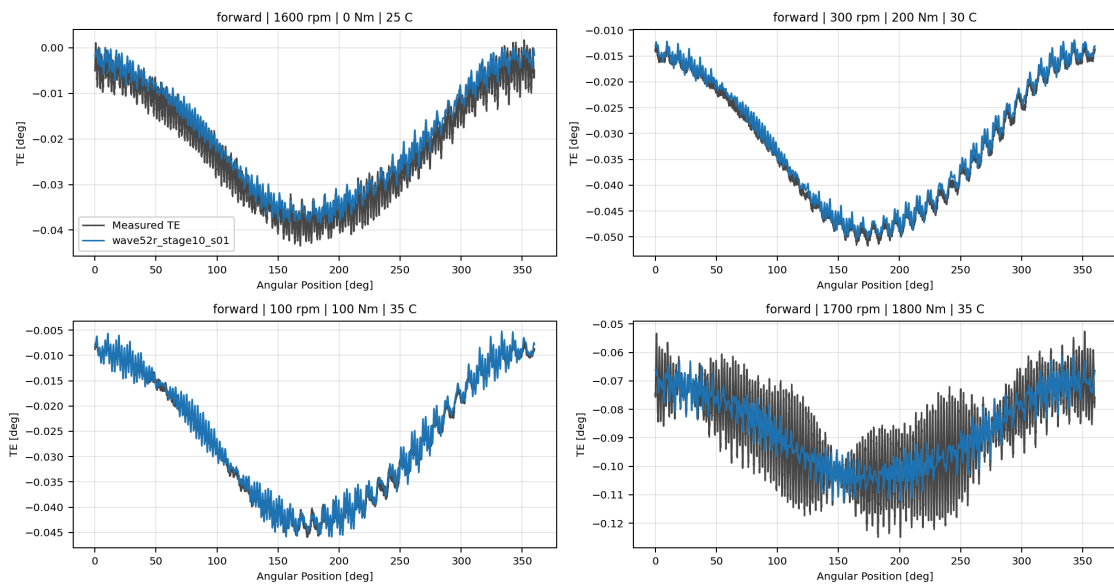
wave52r_stage10_r00



wave52r_stage10_r00 TE Curve Verification Pipeline collage

wave52r_stage10_s01

wave52r_stage10_s01

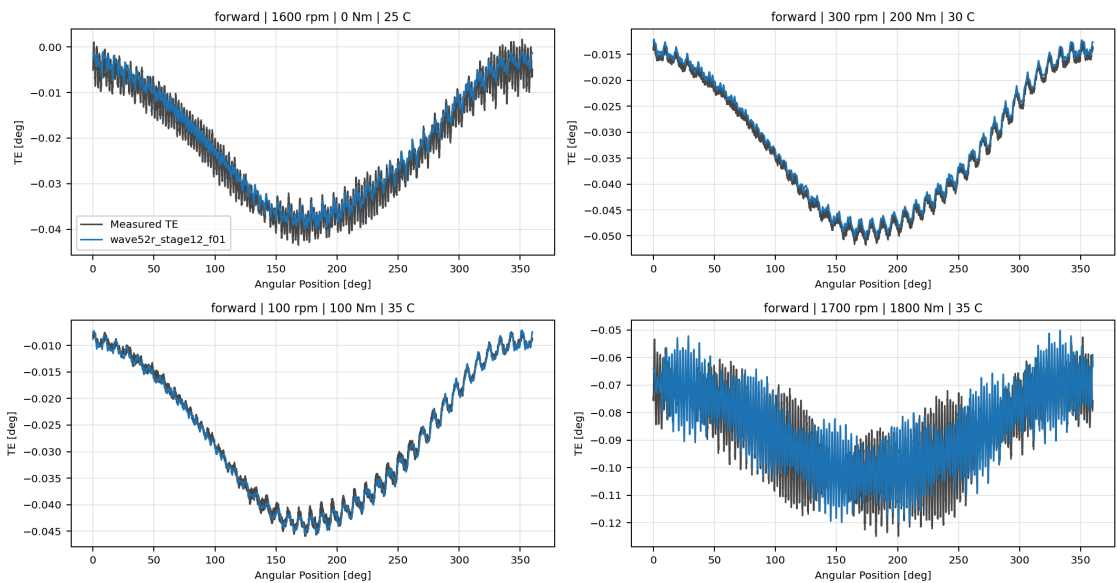


wave52r_stage10_s01 TE Curve Verification Pipeline collage

Collage Gallery - Forward Wave52r Stage12 Temporal Models

wave52r_stage12_f01

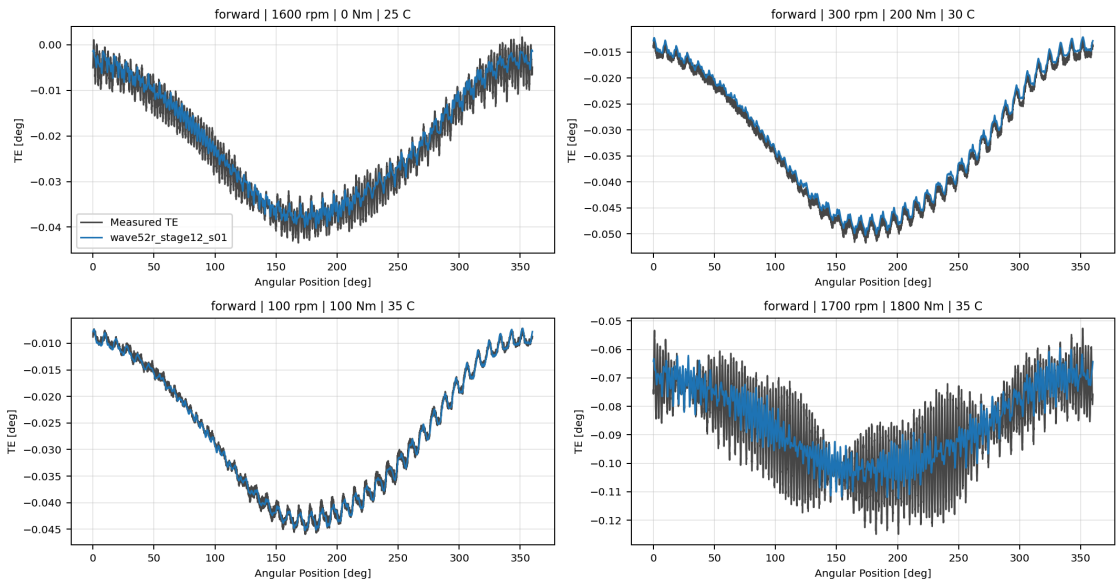
wave52r_stage12_f01



wave52r_stage12_f01 TE Curve Verification Pipeline collage

wave52r_stage12_s01

wave52r_stage12_s01

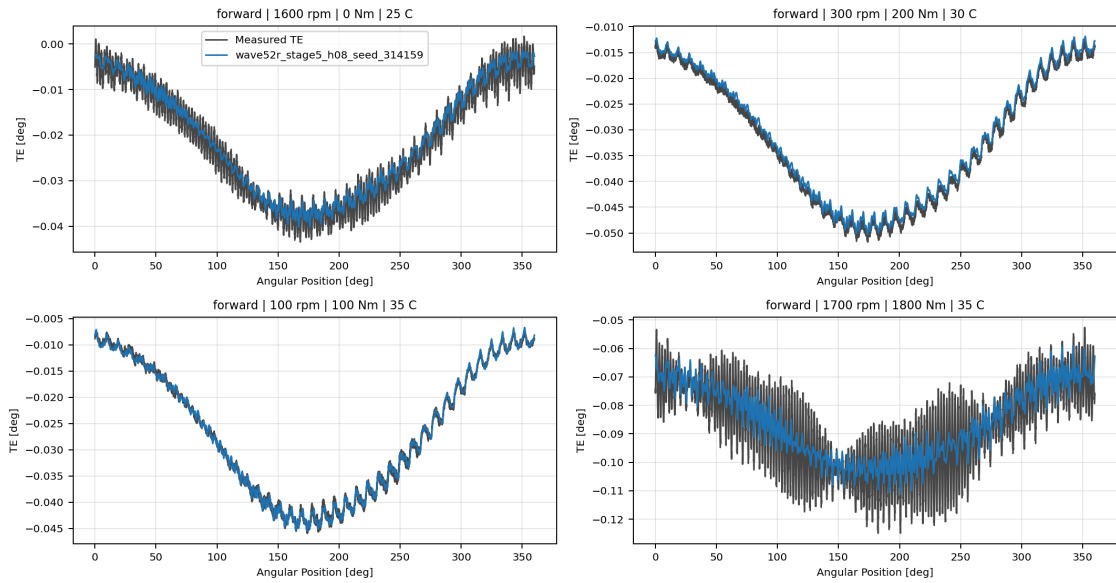


wave52r_stage12_s01 TE Curve Verification Pipeline collage

Collage Gallery - Forward Wave52r Stage5 Non Temporal Models

wave52r_stage5_h08_seed_314159

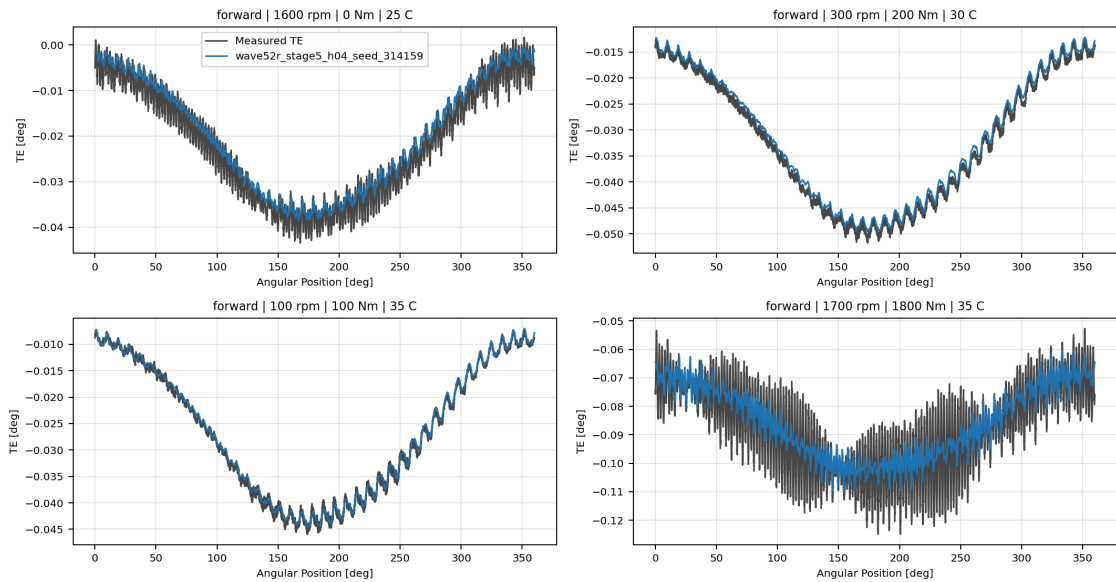
wave52r_stage5_h08_seed_314159



wave52r_stage5_h08_seed_314159 TE Curve Verification Pipeline collage

wave52r_stage5_h04_seed_314159

wave52r_stage5_h04_seed_314159

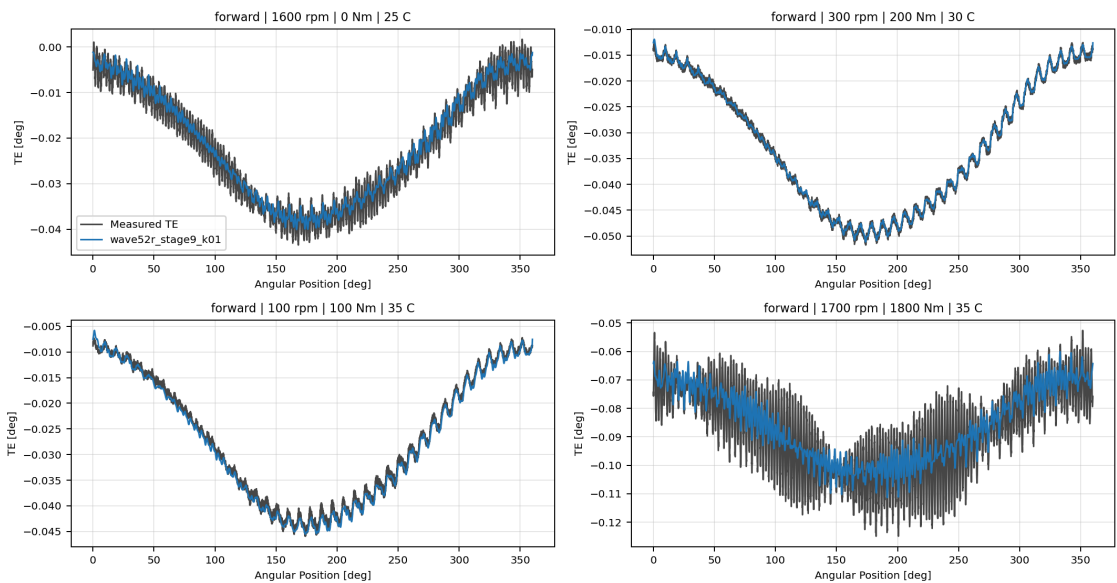


wave52r_stage5_h04_seed_314159 TE Curve Verification Pipeline collage

Collage Gallery - Forward Wave52r Stage9 Temporal Models

wave52r_stage9_k01

wave52r_stage9_k01



wave52r_stage9_k01 TE Curve Verification Pipeline collage

Output Artifacts

- output directory:
output\validation_checks\wave52r_full_candidate_track2_best_model_collage_report\2026-07-30-11-03-21__track2_best_model_collage_report ;
- summary YAML:
output\validation_checks\wave52r_full_candidate_track2_best_model_collage_report\2026-07-30-11-03-21__track2_best_model_collage_report\track2_best_model_collage_summary.yaml ;
- metrics CSV: output\validation_checks\wave52r_full_candidate_track2_best_model_collage_report\2026-07-30-11-03-21__track2_best_model_collage_report\track2_best_model_collage_metrics.csv ;
- report Markdown:
doc\reports\analysis\te_curve_verification_pipeline\02_visual_reports\wave52r_full_candidate_best_model_collage_report\[2026-07-30]\track2_best_model_collage_report.md .